

4 Meselson and Stahl performed experiments that demonstrated semi-conservative replication of DNA.

(a) (i) State what is meant by the term semi-conservative replication.

(2)

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(ii) Name **one** enzyme involved in semi-conservative replication.

(1)

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(iii) Explain the importance of semi-conservative replication in the production of new cells.

(2)

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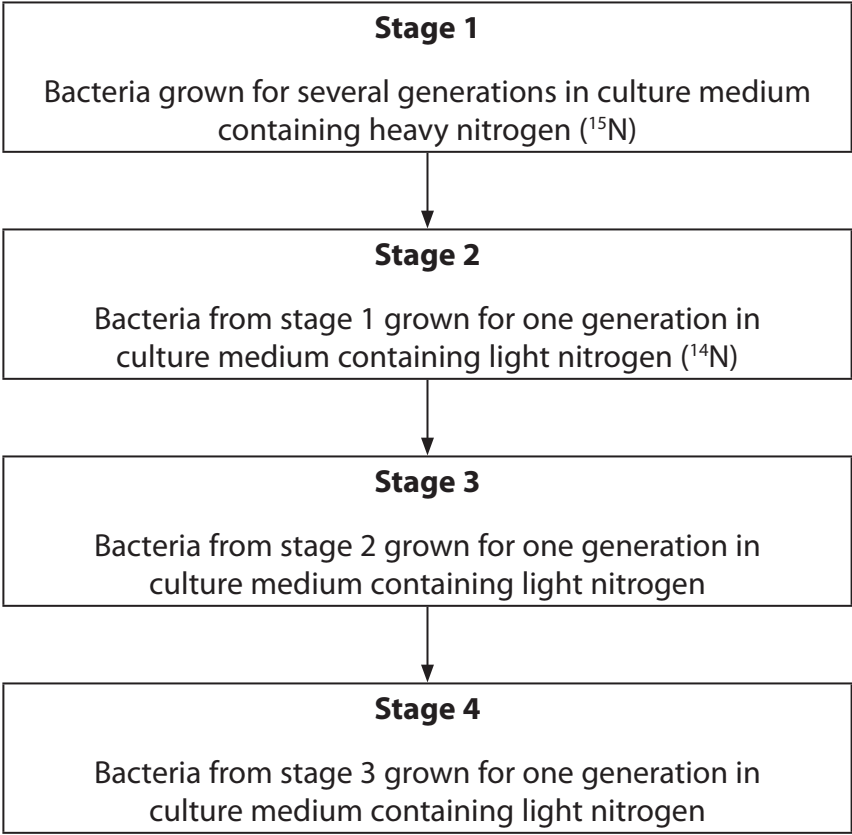
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(b) The flow chart summarises part of one experiment performed by Meselson and Stahl.



Complete the table to show the percentage of the total number of DNA molecules containing heavy nitrogen only, light nitrogen only or both heavy and light nitrogen, at the end of each stage.

(3)

End of stage	Percentage of DNA molecules containing		
	heavy nitrogen only	light nitrogen only	both heavy and light nitrogen
1			
2			
3			

(Total for Question 4 = 8 marks)

